**Minutes of Project meetings:**
Group Members: Jigyasa Tiwari(B23EE1029), Pragati Khandelwal(B23EE1054), Pragati Rokade( B23CM1055), Vandita Gupta(B23CM1061)

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Meeting#1

Date: 01/03/25
Time: 7:30 pm
Venue: Hostel
Attendees: All group members

Discussion points:

- Reviewed the dataset from the provided link to understand its structure and requirements.
- Discussed the project objectives and expected outcomes.

Action item:

- B23CM1055 and B23EE1029 will explore GitHub and determine whether the implementation can be done using Python in VS Code.
- B23EE1054 will analyze the dataset in detail and provide an explanation to all in the next meeting.
- B23CM1061 will research and shortlist suitable algorithms for the project.
- Started reading articles and exploring GitHub repositories to enhance project understanding

Written by: B23EE1029

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Meeting#2

Date: 09/03/25
Time: 11:00 pm
Venue: Hostel
Attendees: All group members

Discussion points:

- Learned how to commit code using VS Code but faced issues loading data.
- Decided to use KNN, Logistic Regression, Naive Bayes, SVM, and Random Forest for classification.
- All members need to learn about Random Forest and SVM in more detail.

Action item:

- B23CM1061 & B23EE1054 will work on implementing Random Forest and Logistic Regression.

- B23EE1029 & B23CM1055 will prepare the Initial Report summarizing the project understanding so far and will look at the feature extraction part.
- B23EE1029 & B23CM1055 will also focus on data preprocessing and implementing KNN and SVM.

Written by: B23EE1029

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Meeting#3

Date: 20/03/25
Time: 8:00 pm
Venue: Hostel
Attendees: All group members

Discussion points:

- Decided to switch to Google Colab for now to streamline implementation.
- Completed preprocessing steps, including denoising, smoothing, sharpening, and filtering images.
- For feature extraction, we have successfully detected shape, edges and corners.

Action item:

- We have successfully completed most of the preprocessing and feature extraction steps.
- Next, we will implement classification models using Naïve Bayes, KNN, and Logistic Regression to evaluate their effectiveness on our dataset.(B23EE1029, B23EE1054, B23CM1061)
- We will also focus on learning the implementation of Random Forest to further enhance our model's performance and will do necessary preprocessing if needed to get better accuracy.
- Will try to finalise the Mid term Report.(B23CM1055)

Written by: B23EE1054

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Meeting#4

Date: 27/03/25
Time: 11:00 pm
Venue: Hostel
Attendees: All group members

Discussion points:

- We have successfully implemented KNN, Naïve Bayes, and Logistic Regression for classification. We will aim to improve accuracy after completing the Random Forest and SVM implementations.
- Random Forest implementation is still in progress, and we aim to complete it by the mid-term report if possible.
- The mid-term report is nearly complete, with only a few refinements remaining.

Action item:

- To further enhance our model, we aim to complete SVM implementation within the next week. (B23CM1061 and B23EE1054)
- Will Work on Random Forest implementation.(B23EE1029 and B23CM1055)

Written by: B23EE1029

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**Meeting#5**

**Date:** 01/04/25
**Time:** 6:00 pm
**Venue:** Library
**Attendees:** All group members

**Discussion points:**

After analyzing multiple projects on GitHub, the team has decided to use HOG (Histogram of Oriented Gradients) and LBP (Local Binary Patterns) for feature extraction.
These methods are selected due to their effectiveness in object detection and low memory consumption, which addresses the performance issues faced on local devices.
A Convolutional Neural Network (CNN) will also be implemented to enhance the accuracy of the existing models.

**Action item:**

- **(B23CM1061 and B23EE1054)** will implement HOG, LBP, and CNN for feature extraction.

- **(B23EE1029 and B23CM1061)** will modify the existing models to integrate the newly implemented feature extraction techniques.

**Written by:** B23EE1054

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Meeting#6

Date: 03/04/25
Time: 10:30 pm
Venue: Hostel
Attendees: All group members

Discussion points:

- Due to frequent **session crashes and RAM limitations on Google Colab**, the team has decided to **shift development to Kaggle** for improved stability and performance.
- Efforts will also begin on preparing the **final project presentation**.

Action item:

- (B23CM1061 and B23EE1054) will work on random Forest Implementation.
- B23EE1029 will do SVM implementation.
- B23CM1055 will explore methods to **improve the accuracy** of the already implemented models, either through **enhanced preprocessing** or by applying **more effective feature extraction techniques**.

Written by: B23EE1029

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Meeting#7

Date: 07/03/25
Time: 6:30 pm
Venue: Hostel
Attendees: All group members

Discussion points:

- We are still facing a lot of issues as the models are crashing due to ram, so we decided to reduce the data size to 10,000 images so that code block could a run a little fast but then thought that if would in return lower the accuracy
- On the implemented models like KNN, random forest we are trying to improve accuracy to a decent level.

Action item:

- B23EE1054 will work on the Report and ppt for now.
- B23EE1029 and B23CM1055 will work on the UI part for the project.
- B23CM1061 will try to work with google cloud.

Written by: B23EE1029

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Meeting#8

Date: 8/03/25
Time: 12:30 am
Venue: Library
Attendees: All group members

Discussion points:
- B23EE1054 and B23EE1029 working on UI and report.
- B23CM1055 and B23CM1061 will try to learn how to integrate Google Cloud..

Action item:
- Will prepare Spotlight Video and ppt for final submission.

Written by: B23EE1029

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Meeting#9

Date: 9/03/25
Time: 7: 00 pm
Venue: Library
Attendees: All group members

Discussion points:
- We are working on UI and report.
- We are done with major part of profile page, some part of UI is left and we are trying to do the models with PCA if that can improve the accuracy of our models.

Action item:
- Will prepare Spotlight Video and whatever is left in the report.
- Update the profile page with results barchart.

Written by: B23EE1029

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Meeting#10

Date: 11/03/25
Time: 12:00 pm
Venue: Hostel
Attendees: All group members

Discussion points:

- We are trying to integrate Google Cloud but are facing some issues.
- We implemented Logistic Regression using PCA and got 45.7 accuracy.

Action item:

- Will prepare Spotlight Video and PPT for final submission.
- We are now implementing every model using PCA.

Written by: B23EE1029

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Meeting#11

Date: 13/03/25
Time: 3:00 pm
Venue: Hostel
Attendees: All group members

Discussion:

We have implemented all the models both with and without PCA. In addition, we have also worked on implementing a Multi-Layer Perceptron (MLP) model to further enhance our project.

Action item:

- We will try to implement the Ensemble method but were facing some issues, so we decided to drop it.
- We are also trying on integrating google cloud.

**Written by:** B23CM1061

Meeting#12

Date: 14/03/25
Time: 3:00 pm
Venue: Hostel
Attendees: All group members

Discussion:

We attempted to implement both the Ensemble method and Google Cloud integration; however, we encountered difficulties in both cases. With the Ensemble method, we faced technical challenges that hindered successful implementation. Additionally, we found the Google Cloud interface to be complex and difficult to navigate, which prevented us from effectively integrating it into our project.

Action item:

- We are now progressing towards the final phase of the project, which involves re-running all the models, thoroughly reviewing the report, and finalizing the video presentation..

**Written by:** B23EE1054